American International University- Bangladesh (AIUB)
Faculty of Engineering
Assignment: 1

Course Name: Introduction to Electrical Circuits **Course Code:** EEE 2101
Faculty Name: [redacted] **Semester: Spring 2024 - 25** **Submission Date: 23/04/24**

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q1	CO1	P.a.1.C3	K1			5	
Q2	CO1	P.a.1.C3	K1			5	
Q3	CO1	P.a.1.C3	K1			5	
Q4	CO2	P.a.2.C3	K2			5	
					Total:	20	

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Note that this problem uses the variables C, D, E, F, G which are the digits of middle of your student ID (AB-CDEFG-H). If any value is zero replace it by (A+B)*H. For example, your Id is 21-78658-2 then A=2, B=1, C=7, D=8, E=6, F=5, G=8, H=2

Instructions:

1. *Use previous page as a cover page for this assignment submission. Different cover page or without cover page, assignment submission will not be acceptable. Include the question in your assignment file.*
2. *Attach the simulation file with the submission.*
3. *No late submission is allowed.*
4. *Copied/identical submissions will be graded as 0 for all parties concerned.*

All Answers at a glance

Question no.	Answers
Problem: 1	(a) For 0°C (32°F), $R = 1.84 \times 10^{-3}\ \Omega$ For 100°C (212°F), $R = 2.63 \times 10^{-3}\ \Omega$
	(b) Change in ~~temperature~~ resistance $= 7.875 \times 10^{-5}\ \Omega$
Problem: 2	I prefer option (b)
	voltage drop, $V = 5.84$ V
Problem: 3	(a) $W = 488160$ kWh
	(b) $t = 1.818$ hours
Problem: 4	(a) $P_i = 1657.78$ W
	(b) $I = 19.38$ A

All solution is explained below with proper application of formula.

Problem-1:

(a) The resistance of a copper wire is 0.002 Ω at room temperature (68°F). What is its resistance at 32°F (freezing) and 212°F (boiling)?

(b) For part (a), determine the change in resistance for each 10° change in temperature between room temperature and 212°F.

(a)

Given,

Resistance, $R_1 = 0.002\ \Omega$

Room temperature, $T_1 = 68°F$

$= \left(\frac{5 \times (F-32)}{9}\right)°C$

$= 20°C$

Let,

$T_2 = 32°F = 0°C$

$T_3 = 212°F = 100°C$

We know,

For copper wire, absolute temperature,

$T = -234.5\ °C$

Now,

For 0°C (32°F)

$$\frac{|T| + T_1}{R_1} = \frac{|T| + R_2 T_2}{R_2}$$

or, $R_2 (|T| + T_1) = R_1 (|T| + T_2)$

or, $R_2 = \frac{R_1 (|T| + T_2)}{(|T| + T_1)}$

or, $R_2 = \frac{0.002 \times (234.5 + 0)}{234.5 + 20}$

$= 1.84 \times_{10}^{-3} \Omega$

For 100°C (212° F),

$$\frac{|T| + T_1}{R_1} = \frac{|T| + T_3}{R_3}$$

$$\Rightarrow R_3 (|T| + T_1) = R_1 (|T| + T_3)$$

$$\text{or, } R_3 = \frac{R_1 (|T| + T_3)}{(|T| + T_1)}$$

$$\text{or, } R_3 = \frac{0.002 (234.5 + 100)}{234.5 + 20}$$

$$= 2.63 \times_{10}^{-3} \Omega$$

$\therefore$ For 0°C (32° F), Resistance will be $= 1.84 \times_{10}^{-3} \Omega$

For 100°C (212°F), Resistance will be $= 2.63 \times_{10}^{-3} \Omega$

(b) Change in Resistance for each 10°

Change in Resistance will be:-

$(2.63 \times 10^{-3} - 0.002)$

$= 6.3 \times 10^{-4}\ \Omega$

Temperature difference;

$\Delta T = T_3 - T_1$

$= (100 - 20)\,°C$

$= 80°C$

Here,

$R_1 = 0.002\ \Omega$

$R_3 = 2.63 \times 10^{-3}\ \Omega$ [From 'a']

Change in Resistance,

$\Delta R = R_3 - R_1$

$T_1 = 68°F = 20°C$

$T_3 = 212°F = 100°C$

Therefore,

For 80°C, change in Resistance $= 6.3 \times 10^{-4}\ \Omega$

$\therefore$ " 1°C " " " $= \left(\frac{6.3 \times 10^{-4}}{80}\right)\Omega$

$= 7.875 \times 10^{-6}\ \Omega$

$\therefore$ " 10°C " " " $= (7.875 \times 10^{-6} \times 10)\ \Omega$

$= 7.875 \times 10^{-5}\ \Omega$

$\therefore$ Change in Resistance for each 10°C change in temperature is $= 7.875 \times 10^{-5}\ \Omega$

Problem-2:

Which would you prefer?

a) A penny for every electron that passes through a wire in 0.01 ms at a current of **D** mA, or

b) A dollar for every electron that passes through a wire in 1.5 ns if the current is **EFG** mA.

c) If a conductor with a current of **CDE** mA passing through it converts **FG** J of electrical energy into heat in 30 s, what is the potential drop across the conductor?

I would prefer option (b). The reasons are as follows:-

(a) A penny for every electron that passes through a wire in 0.01 ms at a current D (6) mA.

For this option,

Current, $I = D = 6mA = 6 \times 10^{-3} A$

Time, $t = 0.01 ms = 0.01 \times 10^{-3} sec$

Now,

Charge, $Q = It$

or, $Q = (6 \times 10^{-3} \times 0.01 \times 10^{-3}) C$

$= 6 \times 10^{-8} C$

We know,

charge of 1 electron $= 1.6 \times 10^{-19} C$

Now,

no. of electrons, $n = \frac{Q}{e}$

or, $n = \frac{6 \times 10^{-8}}{1.6 \times 10^{-19}}$

$= 3.75 \times 10^{11}$ electrons

We also know,

1 penny = 1·10 taka

If 1 electron = 1·10 taka.

$\therefore$ $3.75 \times_{10}{}^{11}$ electron = $(1.10 \times 3.75 \times_{10}{}^{11})$ taka

$= 4.125 \times_{10}{}^{11}$ taka

$\therefore$ For (a); $4.125 \times_{10}{}^{11}$ taka.

Again,

(b) A Dollar for every electrons that passes through a wire in 1·5 ns if current is EFG (599) mA.

For this option,

Current, I = ~~1·599~~ 599 mA

$= 599 \times_{10}{}^{-3}$ A

time, t = 1·5 ns

$= 1.5 \times_{10}{}^{-9}$ sec.

Now, charge,

$Q = It$

or, $Q = (599 \times_{10}{}^{-3} \times 1.5 \times_{10}{}^{-9})$ C

$= 8.985 \times_{10}{}^{-10}$

We know,

charge of 1 electron $= 1.6 \times 10^{-19}$ C

$\therefore$ no. of electron, $n = \frac{q}{e}$

$= \frac{8.985 \times 10^{-10}}{1.6 \times 10^{-19}}$

$= 5.615 \times 10^{9}$ electrons

$\therefore$ For (b); 5.615×10^{9} electrons,

Now, we know,

1 Dollar = 110 Taka.

If 1 electron = 110 Taka

$\therefore 5.615 \times 10^{9}$ electrons $= (110 \times 5.615 \times 10^{9})$ taka

$= 6.17 \times 10^{11}$ taka.

$\therefore$ For (b); 6.17×10^{11} taka.

So, definitely I would prefer (b).

(c)

Given,

Current, $I = 565\ mA = 0.565\ A$ $[\because I = CDE]$

Energy, $W = FG = 99\ W$

time, $t = 30$ sec.

Now,

we have to calculate potential drop (voltage drop) across the conductor.

We know,

$W = V \times I \times t$

or, $V \times I \times t = W$

or, $V = \dfrac{W}{It}$

or, $V = \left(\dfrac{99}{0.565 \times 30}\right) V$

$= 5.84\ V$

Therefore, voltage drop across the conductor, $V = 5.84\ V$

Problem-3:

a) How much energy in kilowatt hours is required to keep a **CDE** W oil-burner motor running 12 h a week for 5 months?

b) How long can a **DEFG** W heater be on before using more than 12 kWh of energy?

(a)

Given,

Power, $P = CDE = 565$ W

time, $t = 12$ h a week for 5 months

1 month = 4 week

$\therefore$ 5 month $= (4 \times 5)$ "

$= 20$ week.

Total time in hours would be:

$t = (12 \times 20)$ hours

$= 240$ hours

Total time in seconds would be:

$t = (240 \times 60 \times 60)$ sec

$\therefore t = 864000$ sec.

Now,

$$W = \frac{Pt}{1000}$$

$$\text{or, } W = \frac{565 \times 864000}{1000}$$

$$= 488160 \text{ KWh}$$

(b)

Given, Power, $P = DEFG = 6599\ W$

Energy, $W = 12\ kwh$

Now,

$$t = \frac{W \times 1000}{P}$$

or, $t = \left(\frac{12 \times 1000}{6599}\right)$ ~~sec~~ hours

$= 1.818$ hours.

Therefore, time, $t = 1.818$ hours

Problem-4:

A motor is rated to deliver 2 hp.

a) If it runs on 110 V and is 90% efficient, how many watts

b) does it draw from the power line?

c) What is the input current?

d) What is the input current if the motor is only 70% efficient?

(a)

Given, motor is rated to deliver 2hp

$1\ hp = 746\ \omega$

$\therefore 2\ hp = (746 \times 2) = 1492\ \omega$

$\therefore P_o = 1492\ \omega$ $[P_o = P_{output}]$

Given, efficiency, $\eta = 90\%$

$= 0.90$

Now,

$$\eta\% = \frac{P_o}{P_i} \times 100\%$$

or, $\eta P_i = P_o$

or, $P_i = \frac{P_o}{\eta}$

or, $P_i = \left(\frac{1492}{0.90}\right) W$

or, $P_i = 1657.78$ W

$\therefore P_i = 1657.78$ W

(b) Yes, it draws from the power line, where, $P_i = 1657.78$ W.

(c) The input current is;

$$I = \frac{P_i}{E}$$

$$\text{or, } I = \left(\frac{1657.78}{110}\right) A$$

$$= 15.07 \text{ A}$$

From 'a' we get,
$P_i = 1657.78$ W
$V = 110$ V

$\therefore$ Input current $= 15.07$ A.

(d) For 70% efficiency,

$$\eta\% = \frac{P_o}{P_i} \times 100\%$$

$$\text{or, } \eta = \frac{P_o}{P_i}$$

$$\text{or, } \eta P_i = P_o$$

$$\text{or, } P_i = \frac{P_o}{\eta}$$

$$\text{or, } P_i = \frac{1492}{0.7}$$

$$= 2131.4 \text{ W}$$

From 'a' we get,
$P_o = 1492$ W
$E = 110$ V
Given,
efficiency, $\eta = 70\%$
$= 0.7$

$$I = \frac{P_i}{E} = \frac{2131.4}{110} = 19.38 \text{ A}$$

$\therefore$ The input current, $I = 19.38$ A.